

## EXERCISES

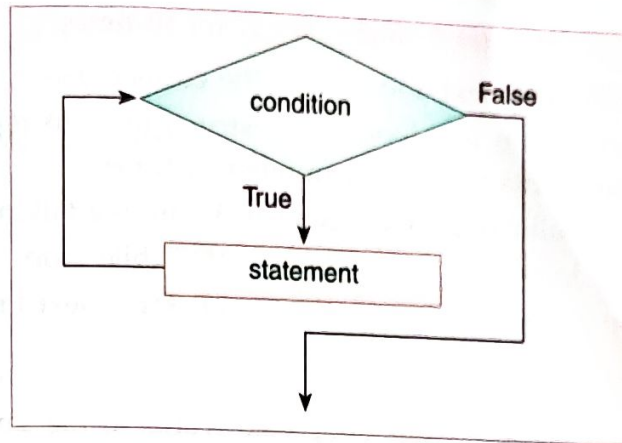


### PART A – COMPETENCY FOCUSED QUESTIONS

#### I. Multiple Choice Questions:

Tick (✓) the correct answer:

1.



Name the type of loop in Java, depicted in the above flowchart.

- (a) while loop (b) do-while loop
  - (c) conditional loop (d) nested loop
2. When execution of the statements are repeated sequentially a number of times in a program, the construct is known as:
- (a) iteration (b) sequence (c) selection (d) none of these
3. Which of the following statement is exit controlled loop?
- (a) for (b) while (c) do-while (d) if-else
4. Which of the following loop execute even once, if the condition is false in the beginning?
- (a) do-while (b) while (c) for (d) nested loop
5. To execute a loop 10 times, which of the following statement will be used?
- (a) for(i=6;i<=26;i=i+2) (b) for(i=3;i<=30;i=i+3)
  - (c) for(i=0;i<10;i++) (d) all of these
6. Which of the following statements creates multiple branching?
- (a) loop (b) continue (c) switch (d) break
7. Which of the following loops checks the condition first and then the execution begins?
- (a) do-while (b) do (c) while loop (d) for
8. To find the sum of whole numbers upto 10, a loop runs:
- (a) once (b) ten times
  - (c) eleven times (d) any number of times
9. How many times will the loop, for (i=1; i++), execute, if there is no statement used to terminate the loop?
- (a) 1 (b) 0 (c) infinite (d) none
10. Which of the following statements uses a case called default?
- (a) do-while (b) while
  - (c) switch (d) all of these

11. A loop statement is given as:

```
for(i=10;i<10;i++)  
{  
    Statement  
}
```

For how many times will the given loop statement be executed?

- (a) none                      (b) 1 time                      (c) 10 times                      (d) infinite

12. Read the following text and choose the correct answer:

Fixed iterative loop is a looping construct, used to repeat the execution of a block of statements for a fixed number of times.

Which of the following loops exclusively are we talking about?

- (a) do-while loop                      (b) while loop  
(c) for loop                      (d) for – next loop

## II. Assertion and Reasoning based questions:

**Assertion (A):** The 'do – while' loop also works in the same way as 'while' loop.

**Reason (R):** Both are considered as 'Exit Controlled' loop.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).  
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).  
(c) Assertion (A) is true and Reason (R) is false.  
(d) Assertion (A) is false and Reason (R) is true.

## III. Application based questions:

A loop based program snippet has a facility to terminate the loop even if the required iterations are not over. You can also create a loop in which the execution repeats for the next iteration leaving the current iteration incomplete. Sometimes, you need to design a loop that has endless number of iterations or without the block of statements whose execution need to be iterated.

Based on the above discussion, answer the following questions:

- (a) Which statement is used to terminate the loop, if its iterations are not over?  
(i) continue                      (ii) break                      (iii) skip                      (iv) jump  
(b) Name the statement that returns to the execution of the next iteration of the loop if current iteration is incomplete.  
(i) continue                      (ii) repeat                      (iii) goto                      (iv) break  
(c) Which of the following is not an endless looping construct?  
(i) for(i = -10; i >= 0; i += 2)                      (ii) for(i = 1; i >= -10; i++)  
(iii) for(i = 0; i <=10;)                      (iv) for( ; )  
(d) A loop without the block of statements is called as:  
(i) Entry controlled loop                      (ii) Exit controlled loop  
(iii) Empty loop                      (iv) Unknown iterative loop

IV. Predict the output of the following program snippets:

1. class dk1

```
{  
    public static void main(String args[ ])  
    {  
        int i;  
        for(i = -1;i<10;i++)  
        {  
            System.out.println(++ i);  
        }  
    }  
}
```

2. class dk2

```
{  
    public static void main(String args[ ])  
    {  
        int i=2,k=1;  
        while (++i<6)  
            k* = i;  
        System.out.println(k);  
    }  
}
```

[ICSE-2010]

3. class dk3

```
{  
    public static void main(String args[ ])  
    {  
        int m=2,n=15;  
        for(int i=1;i<=5;i++)  
        {  
            m++;--n;  
            System.out.println("m="+m);  
            System.out.println("n="+n);  
        }  
    }  
}
```

[ICSE-2010]

4. Determine how many times the body of the loop will be executed.

[ICSE-2010]

class dk4

```
{  
    public static void main(String args[ ])
```

```

    {
        int x=5,y=50;
        while(x<=y)
        {
            y=y/x;
            System.out.println(y);
        }
    }
}

```

## V. Rewrite the following programs:

### 1. Using do while:

```

class Test
{
    public static void main(String args[ ])
    {
        int x,c;
        for(x=10,c=20;c>=10;c=c-2)
        {
            x++;
            System.out.println(x);
        }
    }
}

```

[ICSE-2009]

### 2. Using do while:

```

class Pattern
{
    public static void main(String args[ ])
    {
        int i,j;
        for(i=5;i>=1;i--)
        {
            System.out.print(i);
        }
        System.out.println( );
    }
}

```

[ICSE-2009]

### 3. Using do while:

```

class Number
{

```



```

public static void main(String args[ ])
{
    int i,n=191,c=0;
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
            c=c+1;
    }
    if(c==2)
        System.out.println("Prime");
    else
        System.out.println("Not Prime");
}

```

4. Using while loop:

```

import java.util.*;
public class Number
{
    public static void main(String args[ ])
    {
        int n,r;
        Scanner in = new Scanner(System.in);
        System.out.println("Enter a number: ");
        n=in.nextInt( );
        do
        {
            r=n%10;
            n=n/10;
            System.out.println(r);
        }
        while(n!=0);
    }
}

```

## VI. Unsolved Java Programs:

1. Write the programs in Java to display the first ten terms of the following series:

- (a) 1, 4, 9, 16, .....
- (b) 1, 2, 4, 7, 11, .....
- (c) 3, 6, 9, 12, .....
- (d) 4, 8, 16, 32, .....
- (e) 1.5, 3.0, 4.5, 6.0, .....

- (f) 0, 7, 26, .....  
 (g) 1, 9, 25, 49, .....  
 (h) 4, 16, 36, 64, .....  
 (i) 0, 3, 8, 15, .....  
 (j) 24, 99, 224, 399, .....  
 (k) 2, 5, 10, 17, .....  
 [ICSE-2008]
2. Write a program in Java to input any 50 numbers (including positive and negative). Perform the following tasks:
    - (a) Count and display the positive numbers
    - (b) Count and display the negative numbers
  3. Write a program to calculate and display the sum of all odd numbers and even numbers between a range of numbers from m to n (both inclusive) where  $m < n$ . Input m and n (where  $m < n$ ).
  4. Write a program to enter any 50 numbers and check whether they are divisible by 5 or not. If divisible then perform the following tasks:
    - (a) Display all the numbers ending with the digit 5.
    - (b) Count and display the numbers ending with 0 (zero).
  5. Write a program to display all the numbers between m and n input from the keyboard (where  $m < n$ ,  $m > 0$ ,  $n > 0$ ), check and display the numbers that are perfect square. e.g. 25, 36, 49, ..... are said to be perfect square numbers. [A number is said to be a perfect square if its square root is an integer number.]
  6. Write a program to input marks in English, Maths and Science of 40 students who have passed ICSE Examination 2014. Now, perform the following tasks:
    - (a) Find and display the number of students, who have secured 95% or more in aggregate.
    - (b) Find and display the number of students, who have secured 90% or more in English, Maths and Science.
  7. Write a program in Java to find and display the sum of the following series:
    - (a)  $1 + 4 + 9 + \dots + 400$
    - (b)  $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{20}$
    - (c)  $1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{19}$
    - (d)  $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots + \frac{19}{20}$
    - (e)  $2 - 4 + 6 - 8 + \dots - 20$
    - (f)  $(1*2) + (2*3) + \dots + (19*20)$
 [ICSE-2008]
  8. Write a program to input a number and display the new number after reversing the digits of the original number. The program also displays the absolute difference between the original number and the reversed number.  
**Sample Input:** 194  
**Sample Output:** 491  
 Absolute Difference= 297

9. The Greatest Common Divisor (GCD) of two integers is calculated by the continued division method. Divide the larger number by the smaller, the remainder then divides the previous divisor. The process repeats unless the remainder reaches zero. The last divisor results in GCD. [ICSE-2009]

Write a program to accept two numbers. Find and display their GCD by using the technique discussed above.

**Sample Input:** 45, 20

**Sample Output:** GCD=5

10. Write a program in Java to find and display the sum of the following series:

$$(a) S = a^2 + \frac{a^2}{2} + \frac{a^2}{3} + \dots + \frac{a^2}{10}$$

$$(b) S = a + \frac{a^2}{2} + \frac{a^2}{3} + \dots + \frac{a^{10}}{10}$$

$$(c) S = (a^2) + (a^3) + \dots + (a^{20})$$

$$(d) S = a + a^2 + a^3 + \dots + a^n$$

$$(e) S = 1 + \frac{2^2}{a} + \frac{3^2}{a^2} + \dots \text{ to } n \text{ terms}$$

$$(f) S = \frac{1^2}{a} + \frac{3^2}{a^2} + \frac{5^2}{a^3} + \dots \text{ to } n \text{ terms}$$

$$(g) S = \frac{1}{a} + \frac{1}{a^2} + \frac{1}{a^3} + \dots + \frac{1}{a^n}$$

$$(h) S = \frac{x}{2} + \frac{x}{5} + \frac{x}{8} + \frac{x}{11} + \dots + \frac{x}{20} \quad [\text{ICSE-2008}]$$

11. In order to reach the top of a pole, a monkey in his first attempt reaches a height of 5 feet and in the subsequent jumps, he slips down by 2% of the height attained in the previous jump. The process repeats and finally the monkey reaches the top of the pole. Write a program to input height of the pole. Calculate and display the number of attempts the monkey makes to reach the top of the pole.

12. Write a program to input Principal (p), Rate (r) and Time (t). Calculate and display the amount for first five years which is compounded annually for each year by using the formula:

$$\text{Simple Interest (si)} = \frac{prt}{100}$$

$$p = p + \text{si}$$

[Hint: The amount after each year will be the Principal for the next year.]

13. Write a program to accept a number and check whether it is a 'Spy Number' or not. (A number is a spy if the sum of its digits equals the product of its digits.)

**Sample Input:** 1124

Sum of the digits = 1 + 1 + 2 + 4 = 8

Product of the digits = 1\*1\*2\*4 = 8

14. A tech number has even number of digits. If the number is split in two equal halves, then the square of sum of these halves is equal to the number itself. Write a program to generate and print all four-digit tech numbers. [ICSE 2019]



15. Write a program to input a number and check whether the number is a lead number or not. A lead number is the one whose sum of even digits is equal to the sum of odd digits.

For example,

**Sample Input:** 3669

**Sample Output:** Sum of odd digits =  $3 + 9 = 12$   
Sum of even digits =  $6 + 6 = 12$   
Hence, 3669 is a lead number.

16. Write a program to input a number and check whether it is a twisted prime or not.

A number is said to be a twisted prime, if the number as well as its reverse is also a prime and display an appropriate message accordingly.

**Sample Input:** 37

**Reversed Number:** 73

**Sample Output:** 37 and 73 both are prime. So, it is a 'Twisted Prime'.

17. Write a program to input a three-digit number. Display its digits raised to the power of their respective positions.

**Sample Input:** 465

**Sample Output:**

$$5^1 = 5$$

$$6^2 = 36$$

$$4^3 = 64$$

[Hint: 5 is at Unit's place, 6 is at Ten's place and 4 is at Hundredth's place.]

18. Write a menu driven program to input two positive numbers m and n (where  $m > n$ ) and perform the following tasks:

- Find and display the sum of two numbers without using '+' operator.
- Find and display the product of two numbers without using '\*' operator.
- Find and display the quotient and remainder of two numbers without using '/' or '%' operator.

[Hint: Successive subtraction is known as division, the last value obtained as a result of successive subtractions is the remainder and the number of subtractions will result in the quotient.]

**Sample Input:** m=5, n=2

$$5 - 2 = 3$$

$$3 - 2 = 1, \text{ thus Quotient} = 2 \text{ and Remainder} = 1]$$

19. Write a menu driven program to accept a number from the user. Check and display whether it is a Prime number or an Automorphic number.

- Prime number: (A number is said to be prime, if it is only divisible by 1 and itself)

**Example:** 3, 5, 7, 11, .....

- Automorphic number: (An automorphic number is the number which is contained in the last digit(s) of its square.)

**Example:** 25 is an Automorphic number as its square is 625 and 25 is present as the last two digits.

[ICSE-2010]



20. Write a menu driven program to perform the following tasks by using switch-case statement: [ICSE-2011]

(a) To display the series:

0, 3, 8, 15, 24, ..... to n terms. (Value of 'n' is to be input by the user)

(b) To find and display the sum of the series:

$$S = \frac{1}{2} + \frac{3}{4} + \frac{5}{6} + \frac{7}{8} + \dots + \frac{19}{20}$$

21. Using a switch statement, write a menu driven program to: [ICSE-2012]

(a) Generate and display the first 10 terms of the Fibonacci series

0, 1, 1, 2, 3, 5, .....

The first two Fibonacci numbers are 0 and 1, and each subsequent number is the sum of the previous two numbers.

(b) Find and display the sum of the digits of an integer that is input.

**Sample Input:** 15390

**Sample Output:** Sum of the digits = 18

For an incorrect choice, an appropriate error message should be displayed.

22. Using switch-case statement, write a menu driven program to:

(a) find and display all the factors of a number input by the user ( including 1 and excluding the number itself).

**Sample Input :** n = 15

**Sample Output:** 1, 3, 5

(b) find and display the factorial of a number input by the user (the factorial of a non-negative integer n, denoted by n!, is the product of all integers less than or equal to n.

**Sample Input :** n = 15

**Sample Output:** 5! = 1\*2\*3\*4\*5 = 120

For an incorrect choice, an appropriate error message should be displayed.

[ICSE-2015]

23. Using a switch statement, write a menu driven program for the following:

(a) To find and display the sum of the series given below:

$$S = x^1 - x^2 + x^3 - x^4 + x^5 - \dots - x^{20}; \text{ where } x = 2$$

(b) To display the series:

1, 11, 111, 1111, 11111

[ICSE-2017]

For an incorrect option, an appropriate error message should be displayed.

## PART B – SUBJECTIVE QUESTIONS

### I. Answer the following questions:

1. What do you understand by iterative process? How can it be resolved by using loop?
2. Explain a for loop with an example.
3. Name the different types of looping constructs.
4. What are the parameters needed to create a for loop?
5. Define the following with their constructs:  
(a) Entry controlled loop (b) Exit controlled loop
6. Write down the general format of:  
(a) for loop (b) do – while (c) while loop
7. What is the purpose of using:  
(a) break statement (b) continue statement in a program?
8. What do you understand by inter-conversion of loops?
9. What are the different ways to inter-convert the loops? Name them.
10. Define the following:  
(a) Finite loop (b) Delay loop  
(c) Infinite loop (d) Null loop
11. Distinguish between:  
(a) for loop and while loop (b) while loop and do-while loop
12. State one difference and one similarity between while and do-while loop.
13. State one similarity and one difference between while and for loop.
14. Give one difference between Step loop and Continuous loop.
15. Name two jump statements and their uses.